National University of Computer and Emerging Sciences

Lab Manual 05

Programming Fundamentals

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Activities:

Single-Level do-while Loop

1. **Discount Calculation Based on Price** Write a program that prompts the user to enter the price of an item. Use a do-while loop to validate the input (the price must be a positive number). Once a valid price is entered, apply a discount based on the price and display the final price:
   - Price below $100: 5% discount
   - Price between $100 and $500: 10% discount
   - Price above $500: 15% discount
   - The program should continue until the user enters -1 to quit.

   Example Input: 250
   Output: Final price after discount = $225
2. **Divisibility Check:** Write a program that reads integers from the user and counts how many of them are divisible by both 3 and 5. The loop should stop when the user enters -999.
   Example Input: 15 30 45 60 -999
   Output: Numbers divisible by 3 and 5 = 4

Single-Level for Loop

3. **Factorial Calculation** Write a program that calculates the factorial of a given number N using a for loop. Use a do-while loop to validate that N is a non-negative integer.
   Example Input: N = 5
   Output: Factorial of 5 = 120
4. **Sum of Odd Numbers** Write a program that computes the sum of all odd numbers between 1 and a user-defined number N. Use a for loop to iterate through the numbers and add the odd ones. Ensure that N is a positive integer using a do-while loop.
   Example Input: N = 10
   Output: Sum of odd numbers = 25
5. **Fibonacci Sequence:** Write a program that prints the first N numbers of the Fibonacci sequence. N is provided by the user. Ensure that user N is a positive integer using input validation.
   Example Input: 6
   Output: 0 1 1 2 3 5
6. **Counting Digits of a Number** Write a program that counts the number of digits in a user-specified number N using a for loop. In your program, you have to ensure that N is not a negative number.
   Example Input: N = 54321
   Output: Number of digits = 5